Introduction to Apache Airflow

INTRODUCTION TO APACHE AIRFLOW IN PYTHON



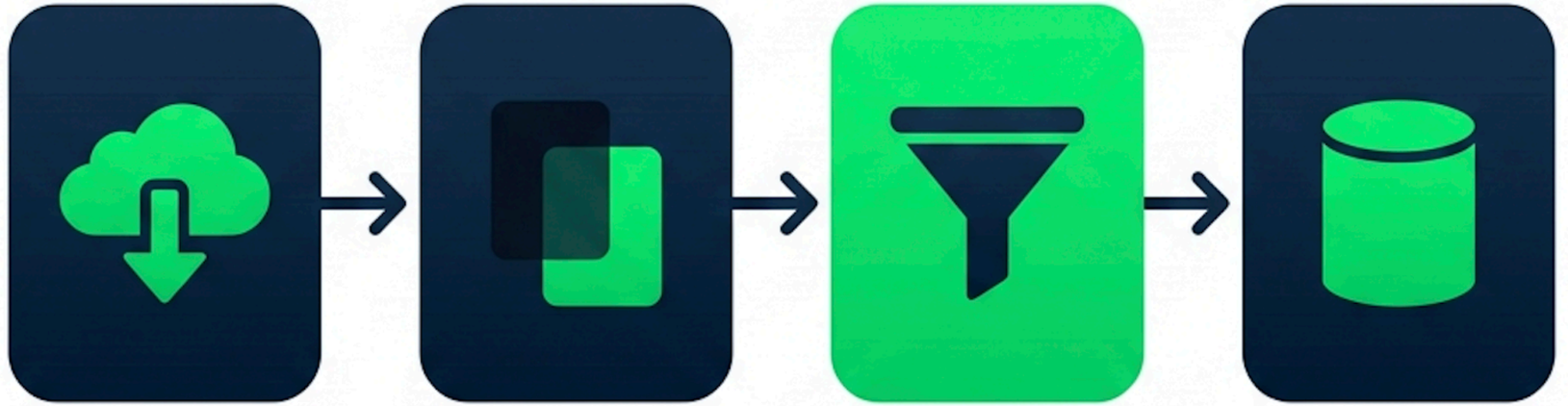
Mike Metzger
Data Engineer

datacamp

What is a workflow?

- **Workflow** - A set of steps to accomplish a given data engineering process

- Example: Download files, copy data, filter information, and write to a database



What is Airflow?

Airflow - a platform to orchestrate workflows:

- Create
- Schedule
- Monitor





What is Airflow?

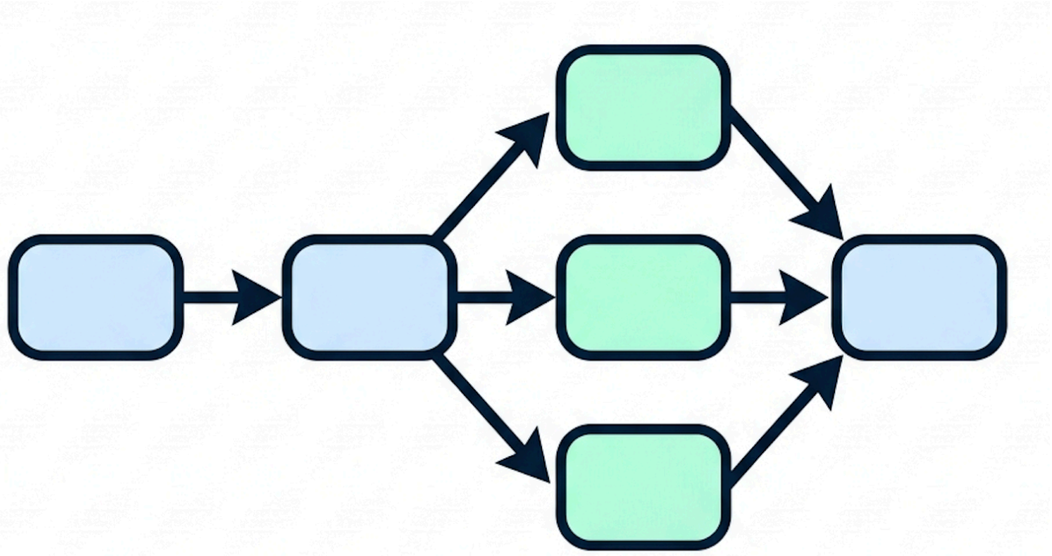
- Can implement programs from any language, but workflows are written in Python
- Implements workflows as Dags
- Accessed via web interface, code, command-line, or via REST API
- Used for ETL pipelines, ML workflows, automation, etc



Quick introduction to Dags

Dag - a model that represents everything needed to execute a workflow.

- Consists of the tasks and their dependencies
- Created with various details including the name, email, owner

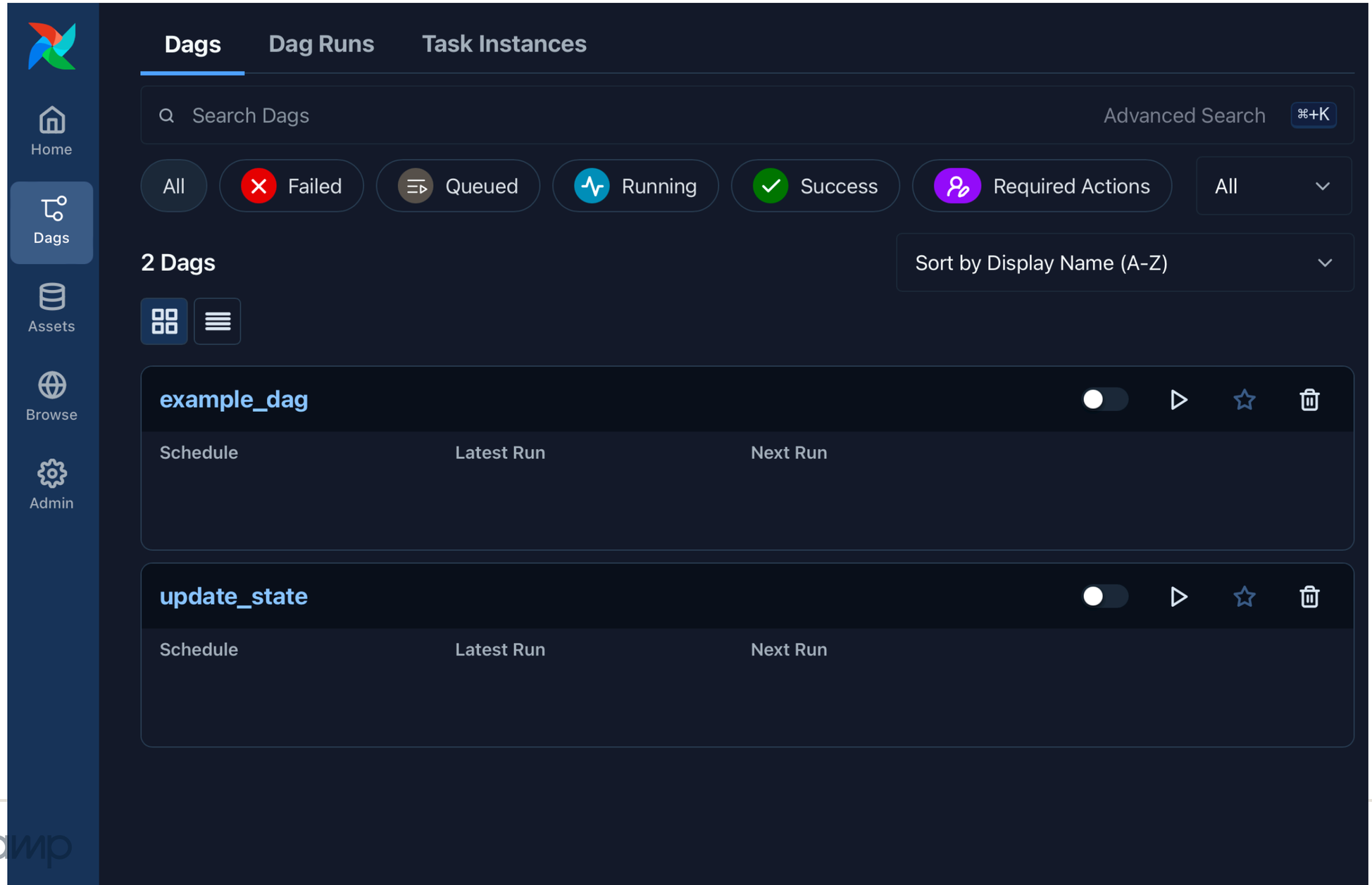


Airflow components

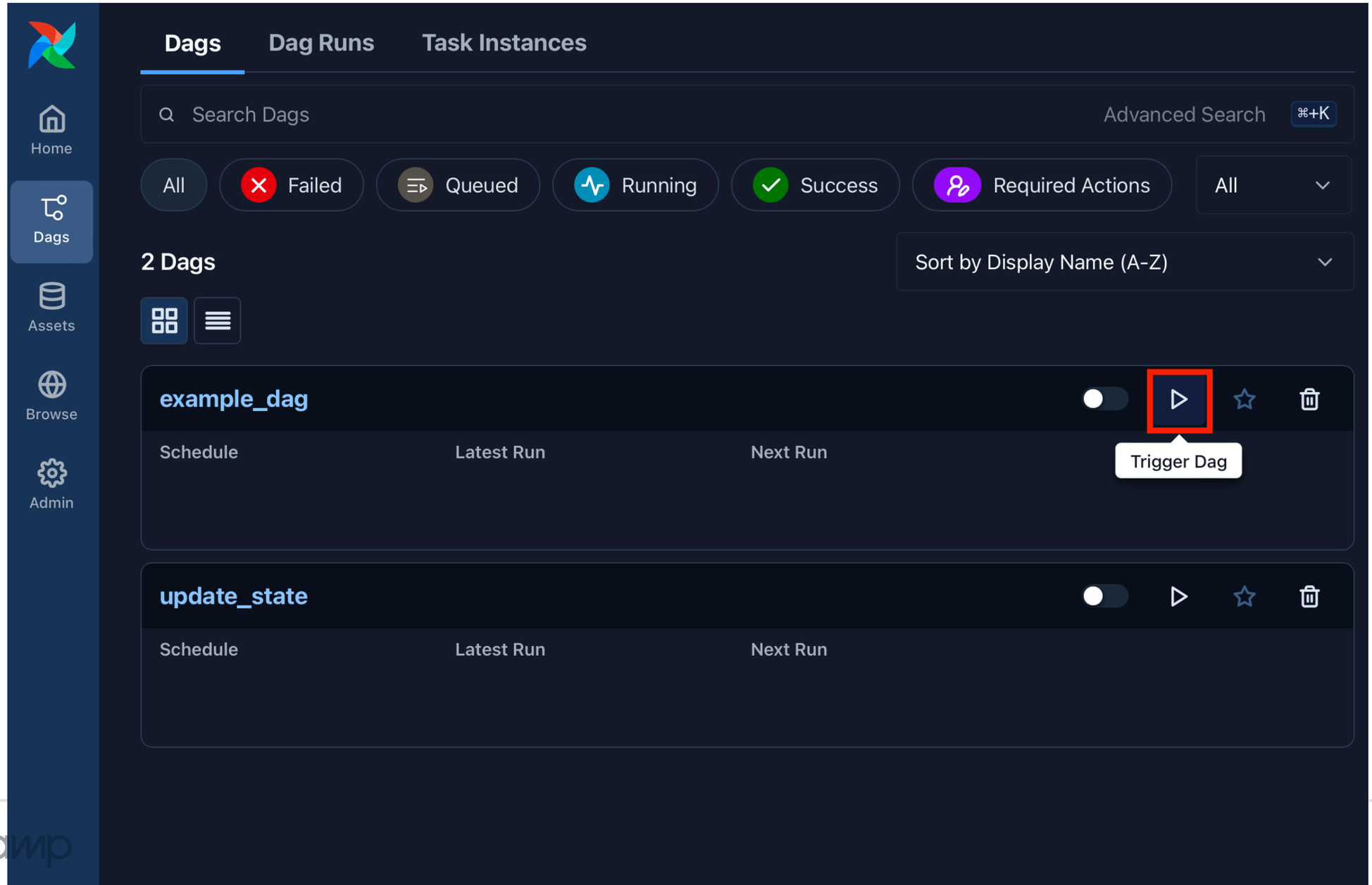
- *Scheduler* -Triggers scheduled workflows, submits tasks
- *API Server* - Provides consistent secure access to the Airflow platform
- *Dag processor* - Read by the scheduler to figure out what tasks to run and when to run them
- *Metadata database* - Stores the state of Dags and tasks



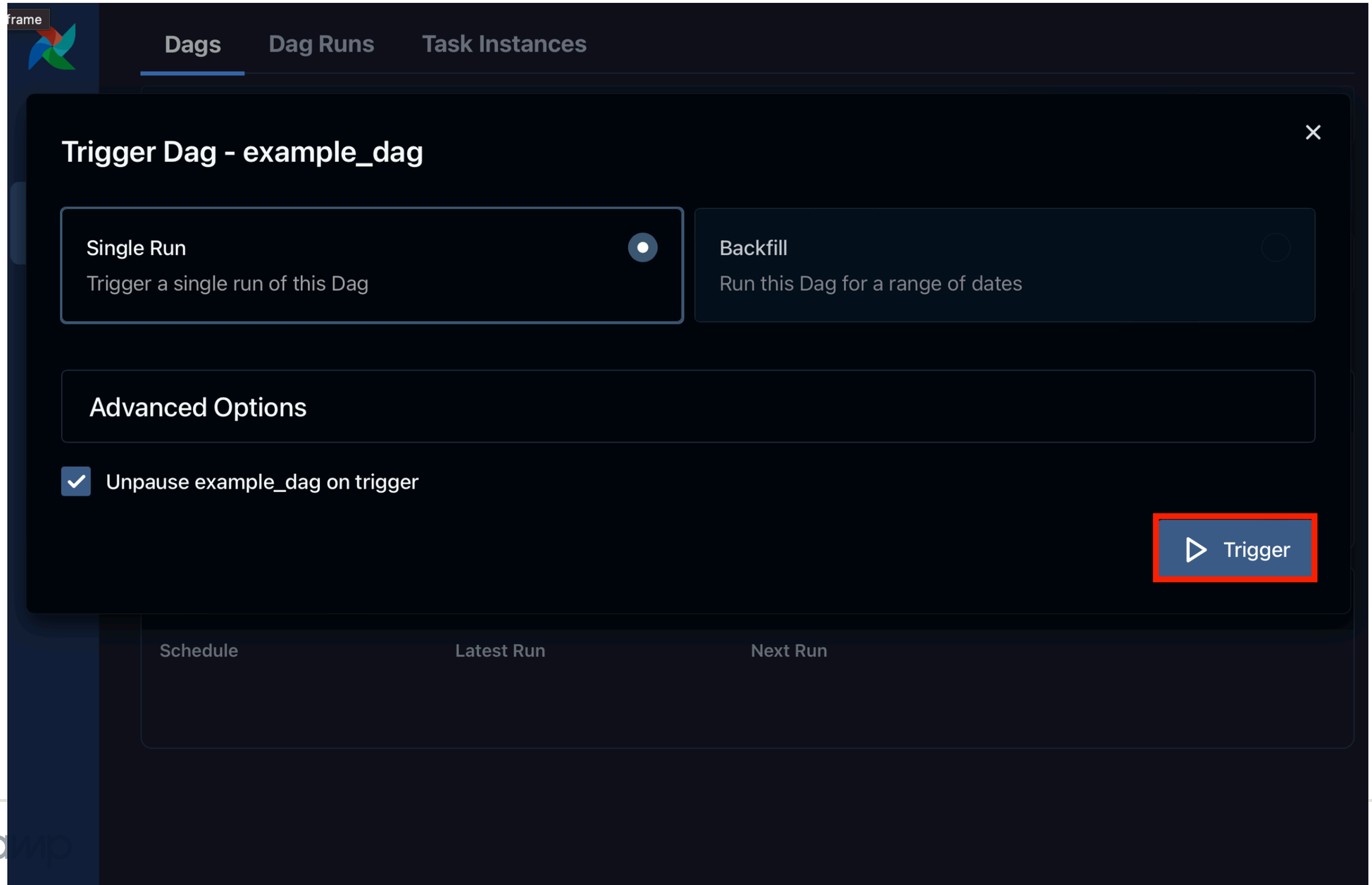
Running a workflow in Airflow UI



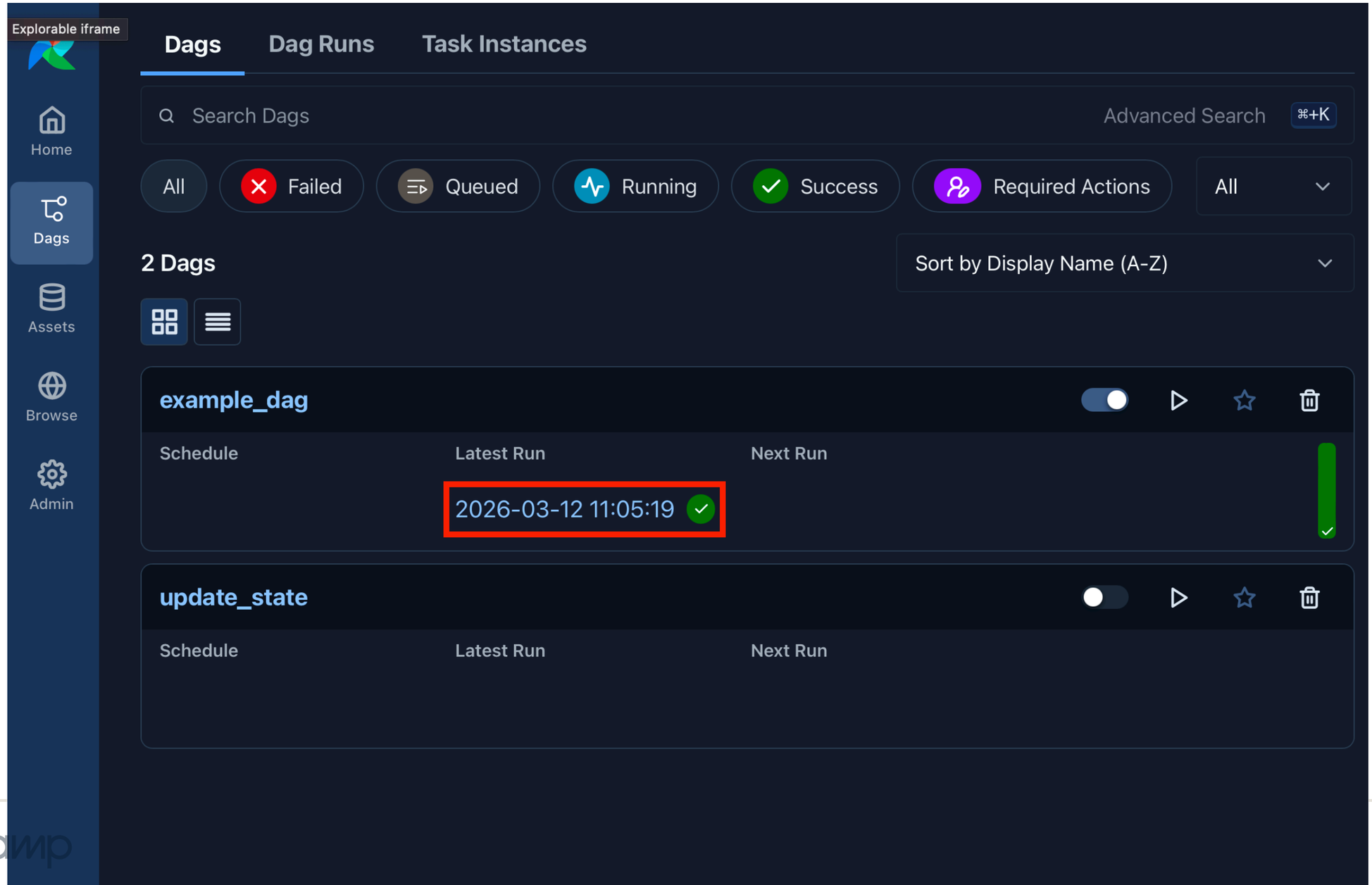
Running a workflow in Airflow UI



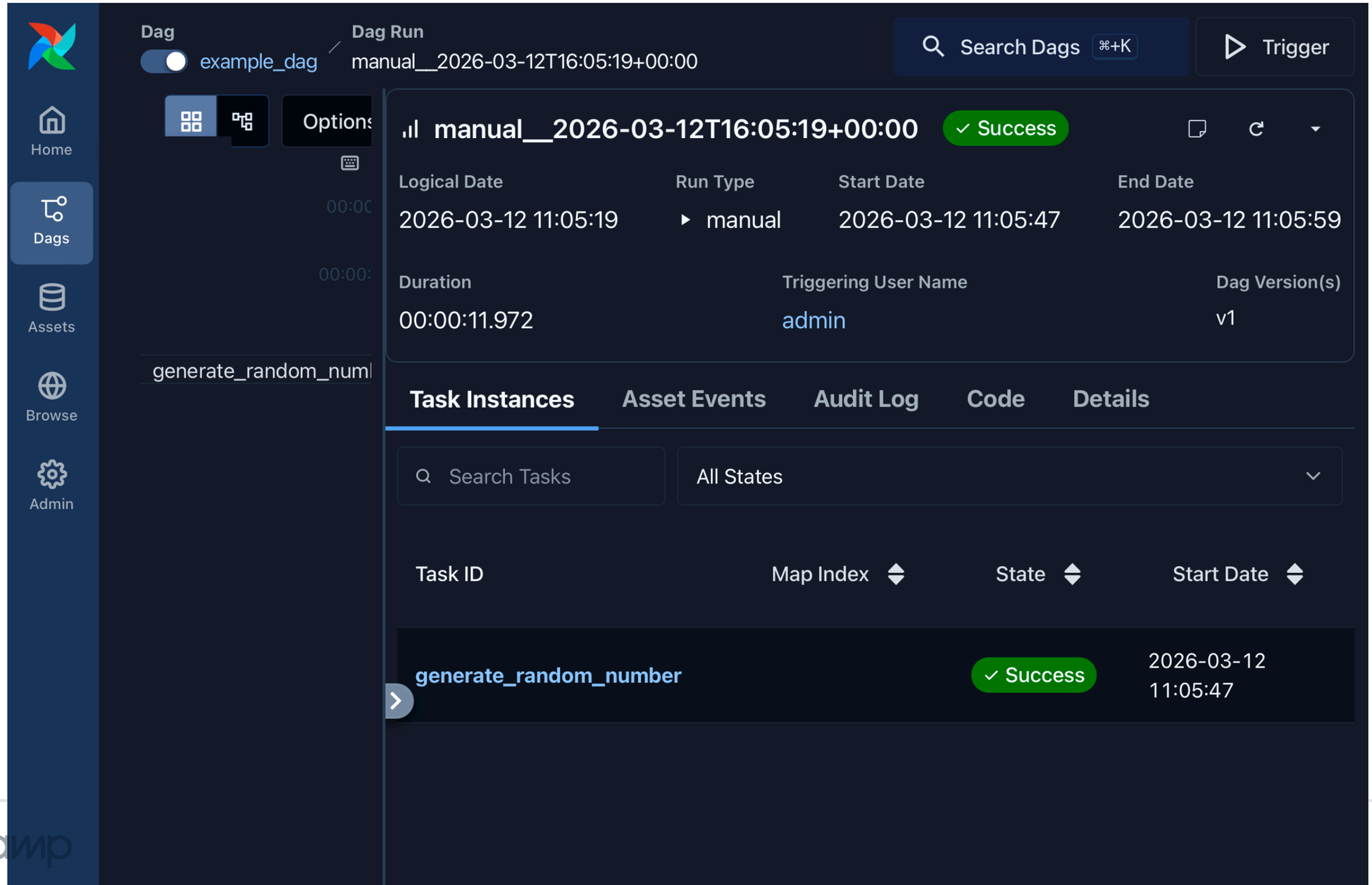
Running a workflow in Airflow UI



Running a workflow in Airflow UI



Running a workflow in Airflow UI



Let's practice!

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Airflow Dags

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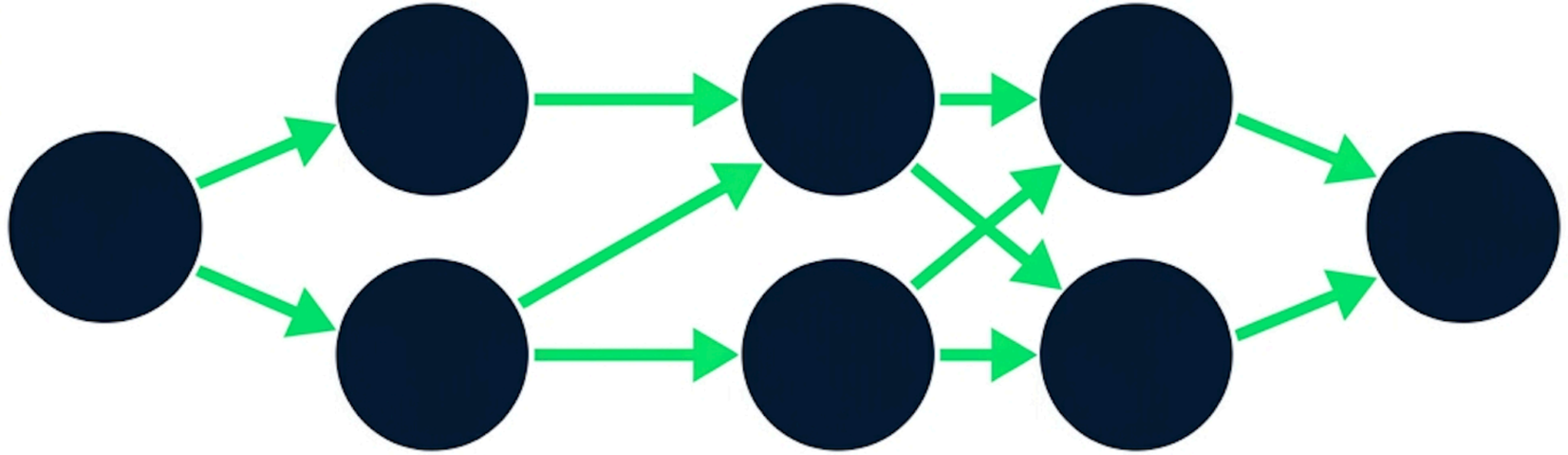
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What is a Dag?

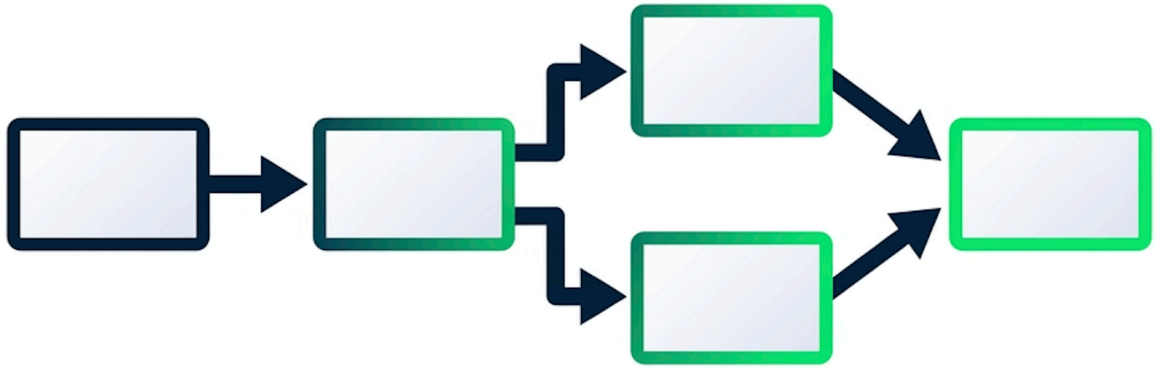
Dag, or *Directed Acyclic Graph*:

- *Directed* - flow between components represents dependencies
- *Acyclic* - does not loop or repeat
- *Graph* - the set of components



Dag in Airflow

- Written in Python (but can use components written in other languages)
- Made up of Tasks to be executed, such as operators or sensors
- Contain dependencies defined explicitly or implicitly
  - ie, Copy the file to the server before trying to import it to the database service.



Define a Dag

Example Dag (Taskflow API):

```
from airflow.sdk import dag


from pendulum import datetime
@dag(
  dag_id='etl_workflow',
  email='jdoe@datacamp.com',
  start_date=datetime(2026, 3, 15, tz="UTC")
)
def etl_workflow():
  ...
etl_workflow()
```

Dags on the command line

- `airflow` command line contains many subcommands
- `airflow -h` - help and subcommand descriptions

- Dag subcommands
  - `airflow dags list` - show all recognized Dags
  - `airflow dags reserialize` - force Airflow to reload Dag files
  - `airflow tasks test` - run a specific task

Command line vs Python

Use the command line tool to:

- Start Airflow processes
- Manually run Dags / Tasks
- Get logging information from Airflow



Use Python to:

- Create a Dag
- Edit the individual properties of a Dag



Let's practice!

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Airflow web interface

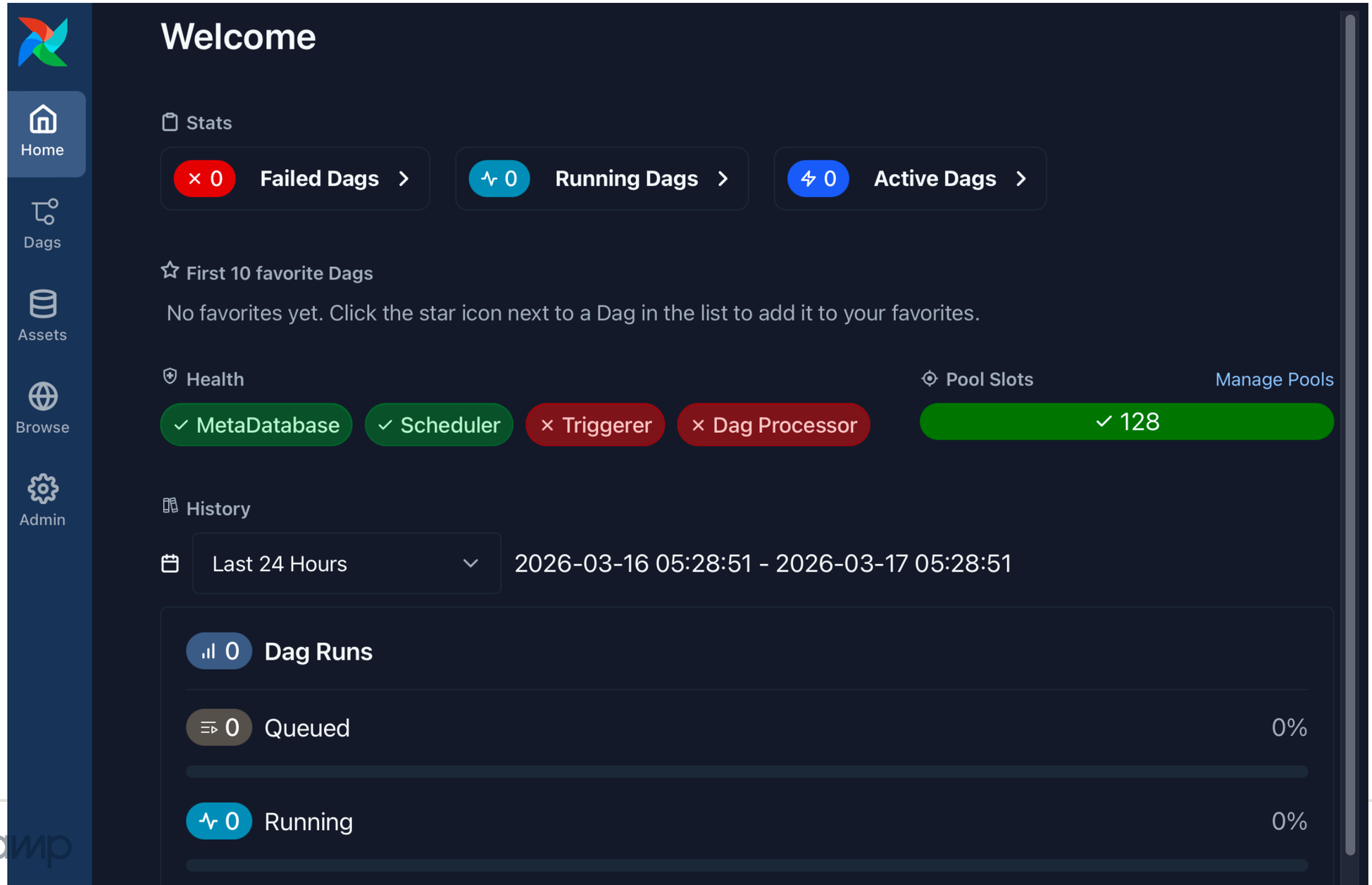
INTRODUCTION TO APACHE AIRFLOW IN PYTHON



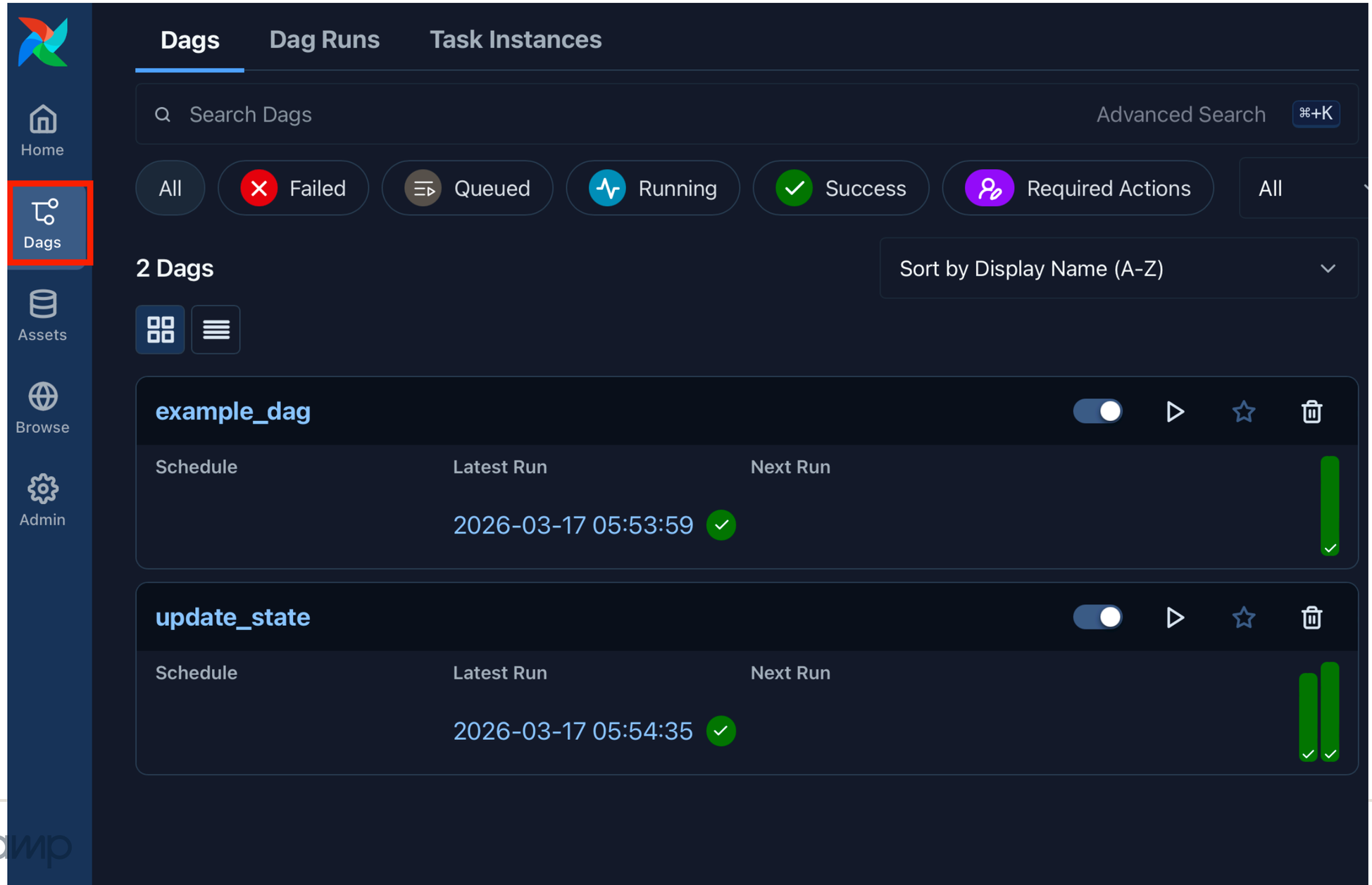
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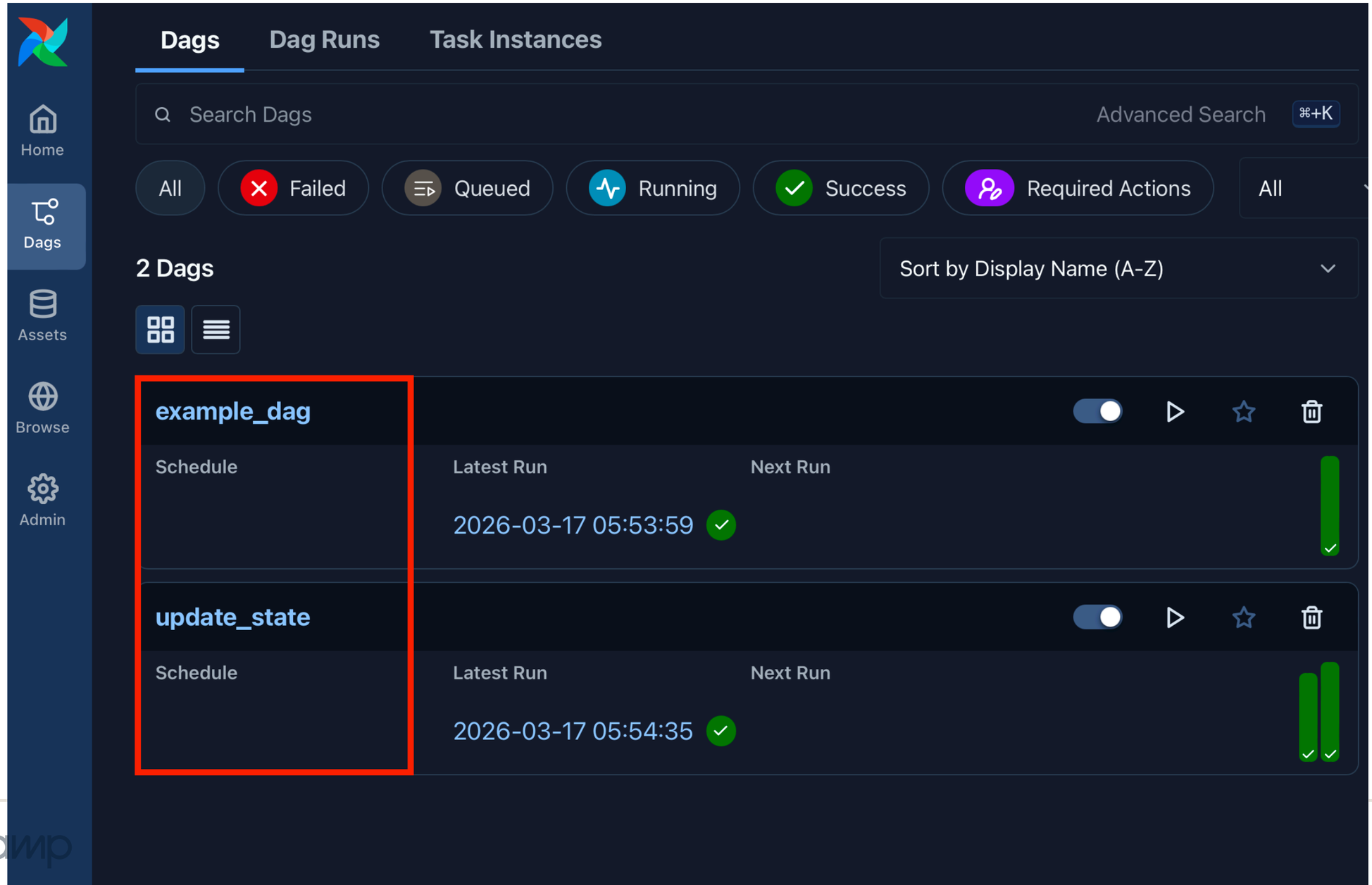
Home View


Welcome
Home
Dags
Assets
Browse
Admin
Stats
0 Failed Dags
0 Running Dags
0 Active Dags
First 10 favorite Dags
No favorites yet. Click the star icon next to a Dag in the list to add it to your favorites.
Health
MetaDatabase
Scheduler
Triggerer
Dag Processor
Pool Slots
Manage Pools
128
History
Last 24 Hours
2026-03-16 05:28:51 - 2026-03-17 05:28:51
0 Dag Runs
0 Queued
0%
0 Running
0%

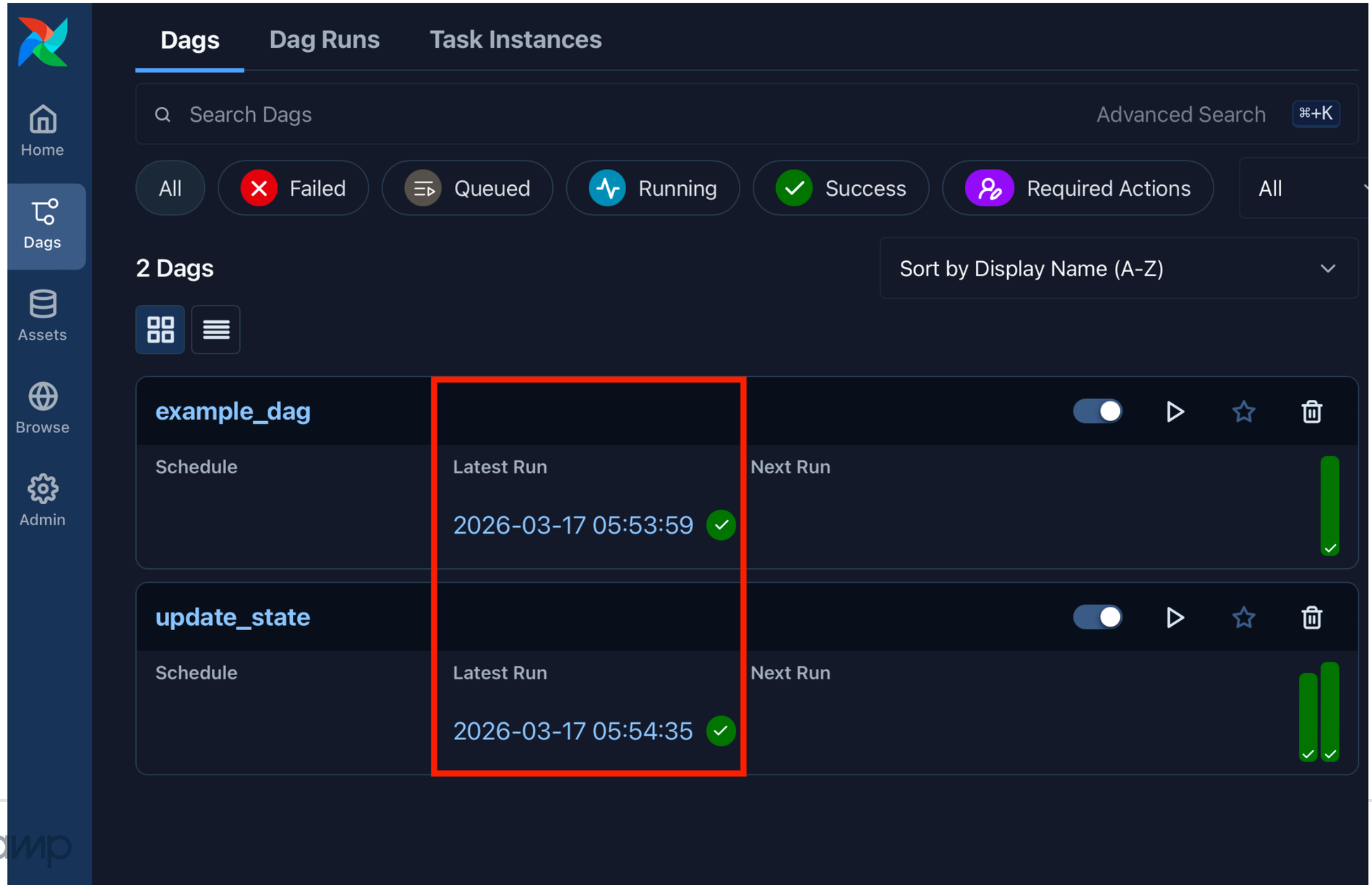
Dags view


Dags
Dag Runs
Task Instances
Search Dags
Advanced Search
⌘+K
Home
Dags
Assets
Browse
Admin
All
Failed
Queued
Running
Success
Required Actions
All
2 Dags
Sort by Display Name (A-Z)
example_dag
Schedule
Latest Run
Next Run
2026-03-17 05:53:59
update_state
Schedule
Latest Run
Next Run
2026-03-17 05:54:35

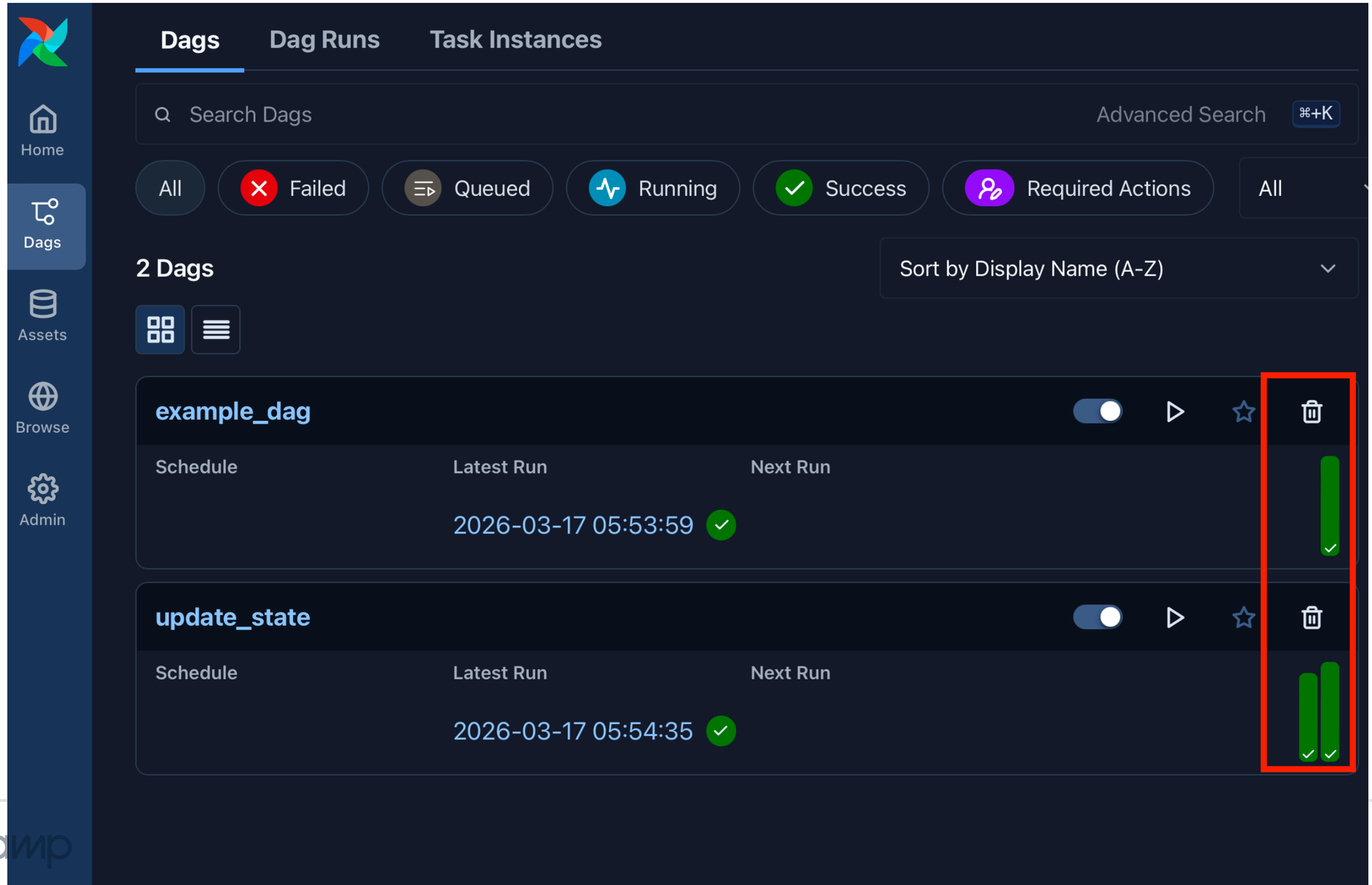
Dags view Dags



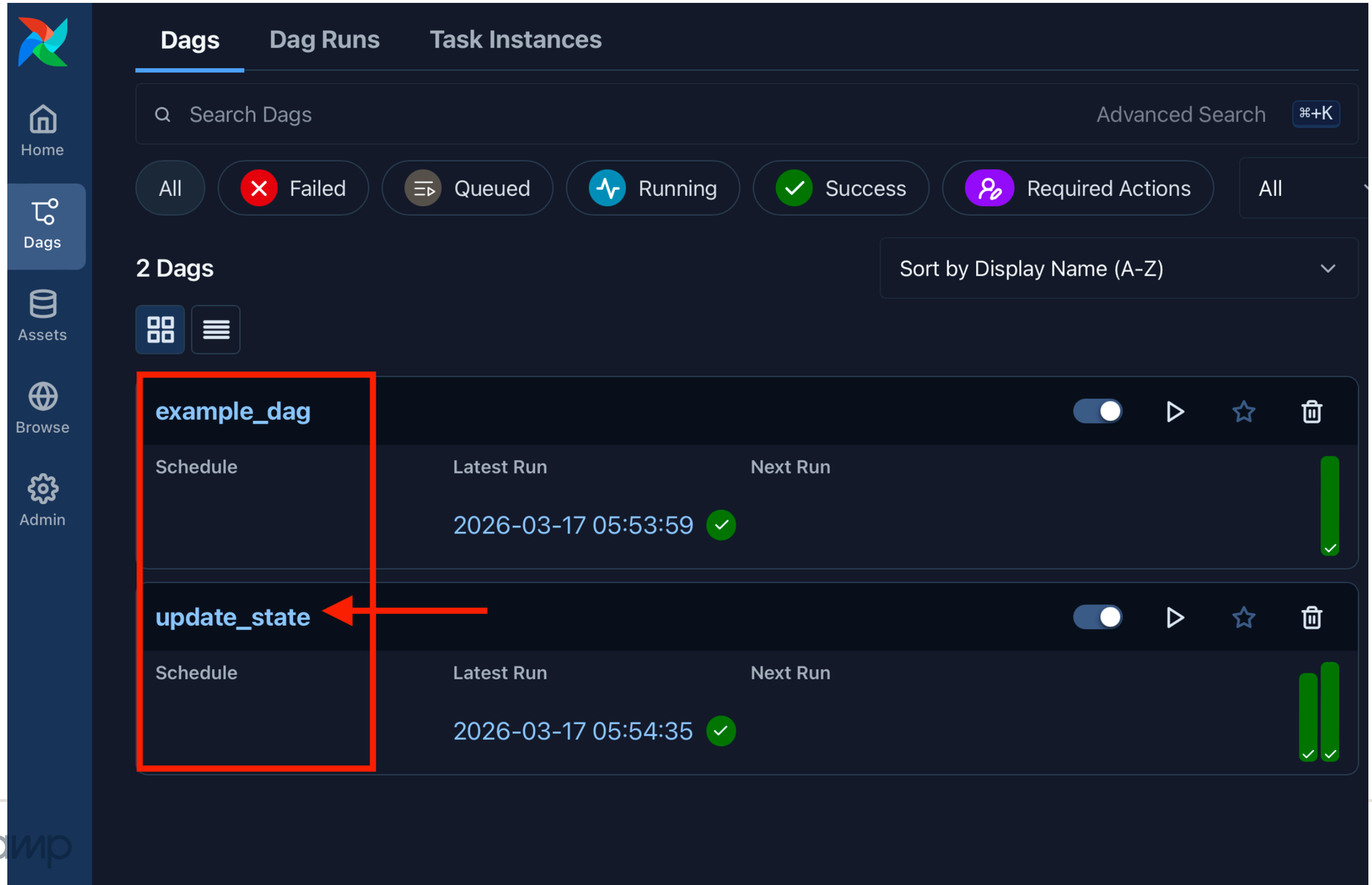
Dags view Latest Run



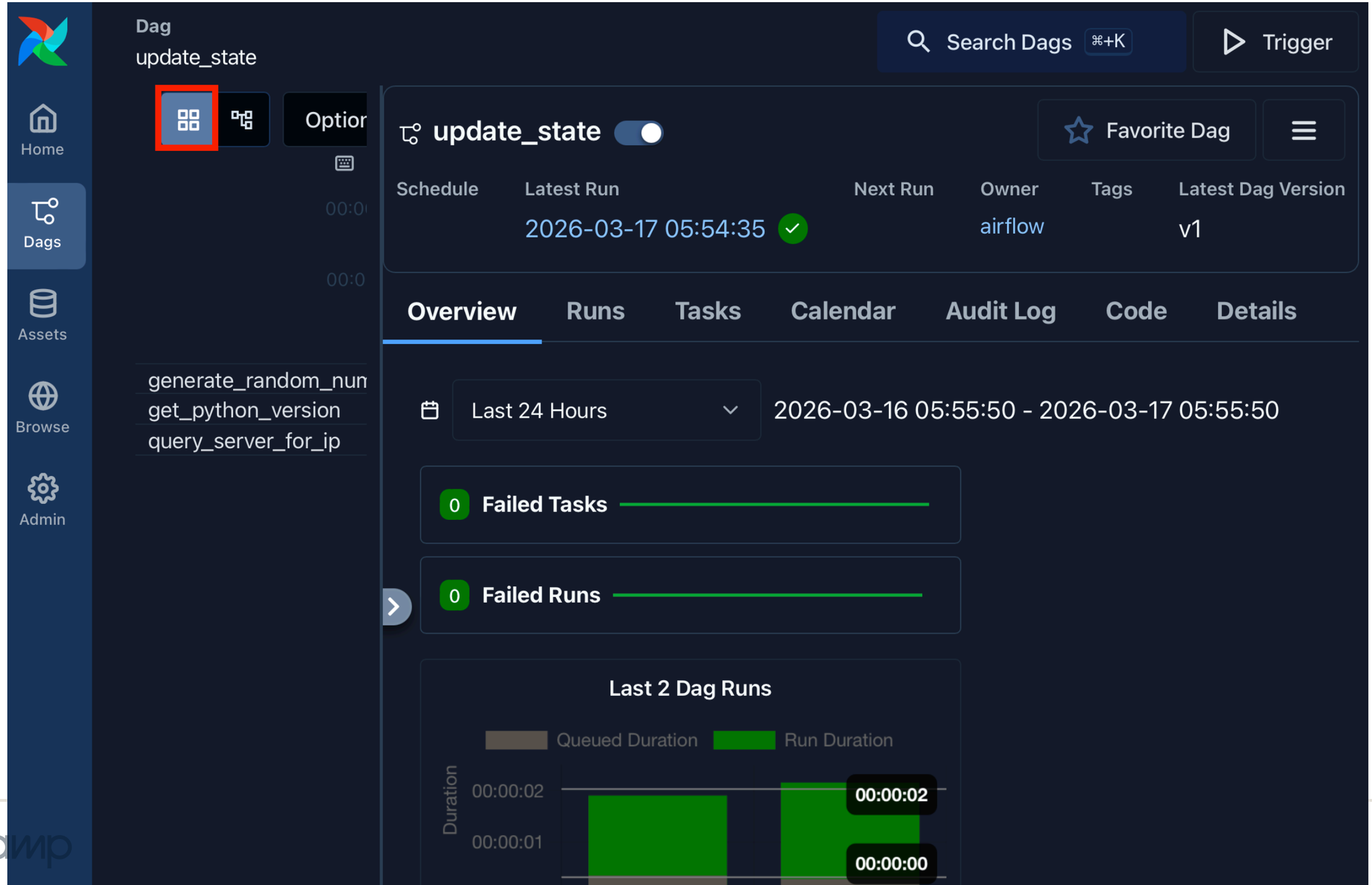
Dags view runs



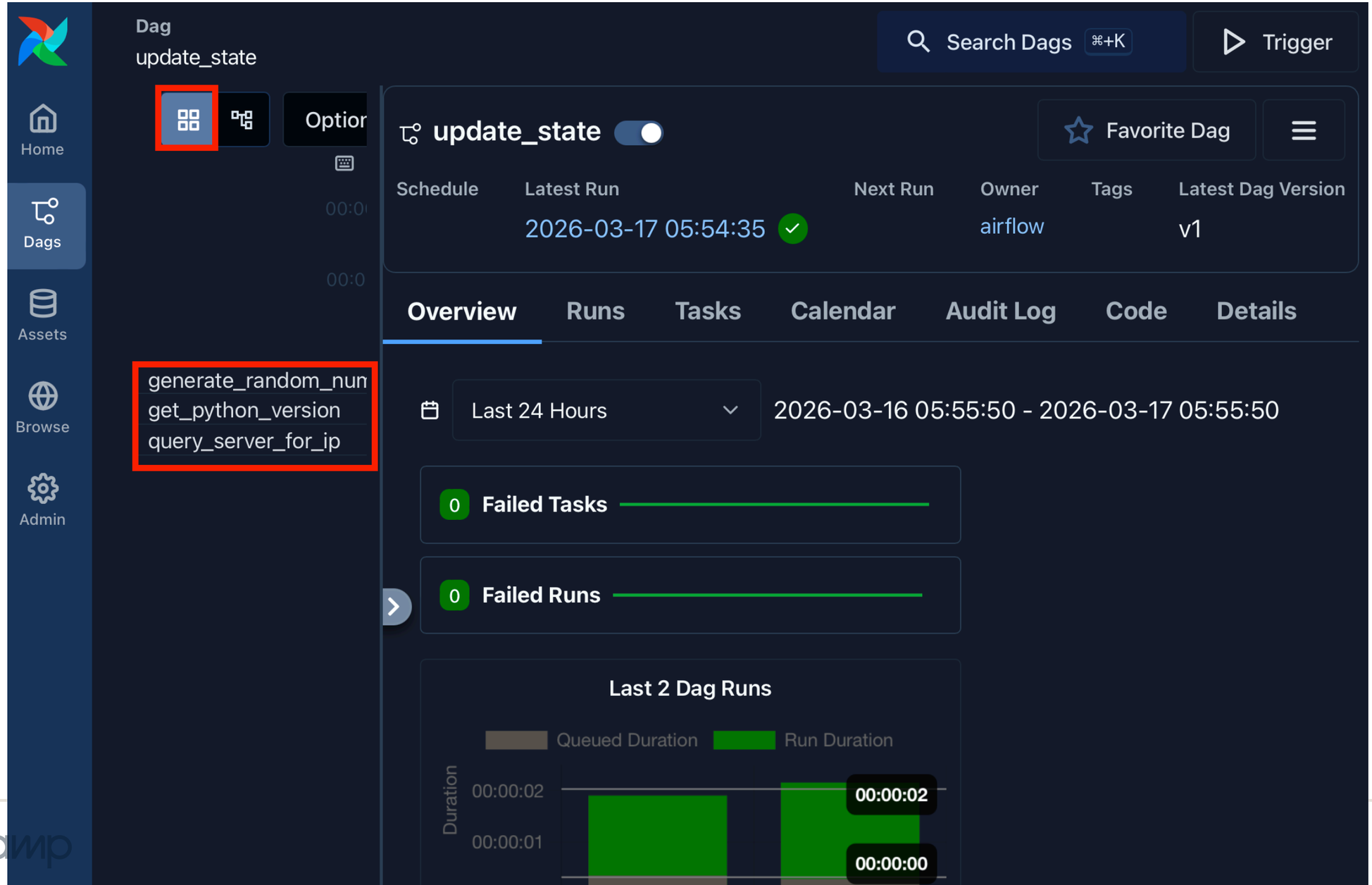
Dag detail navigation



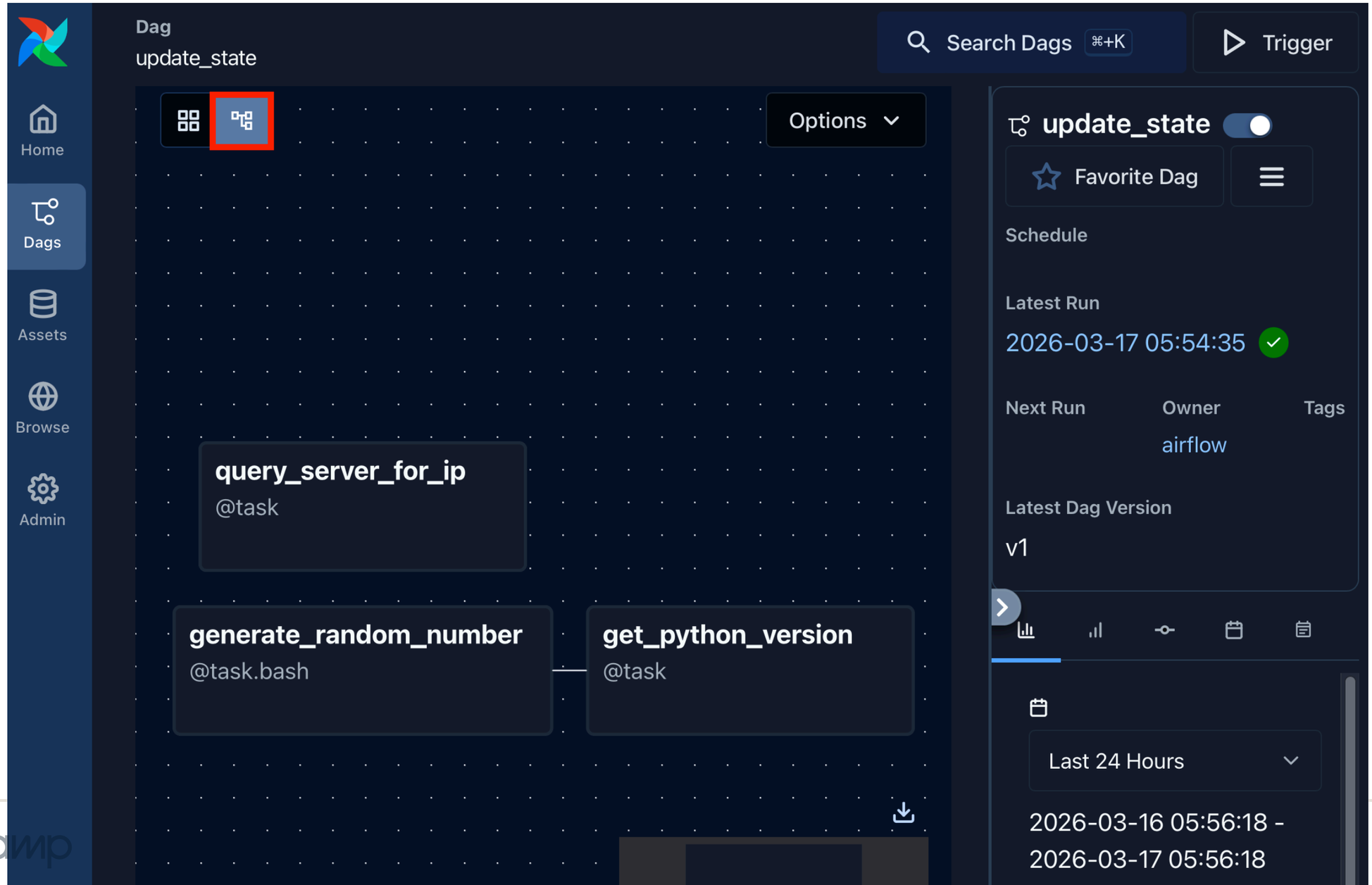
Dags view update_state (Grid)



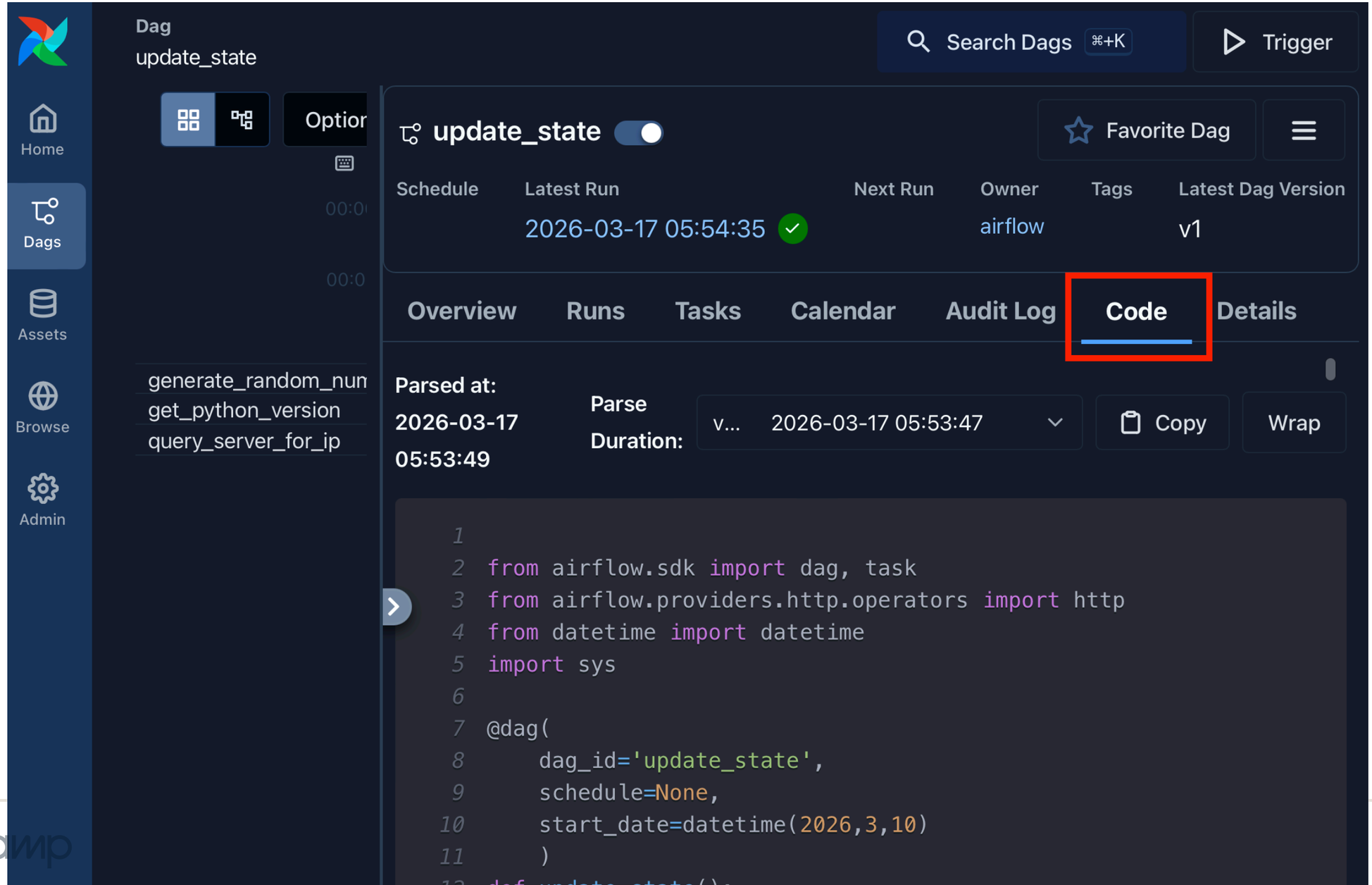
Dags view update_state tasks



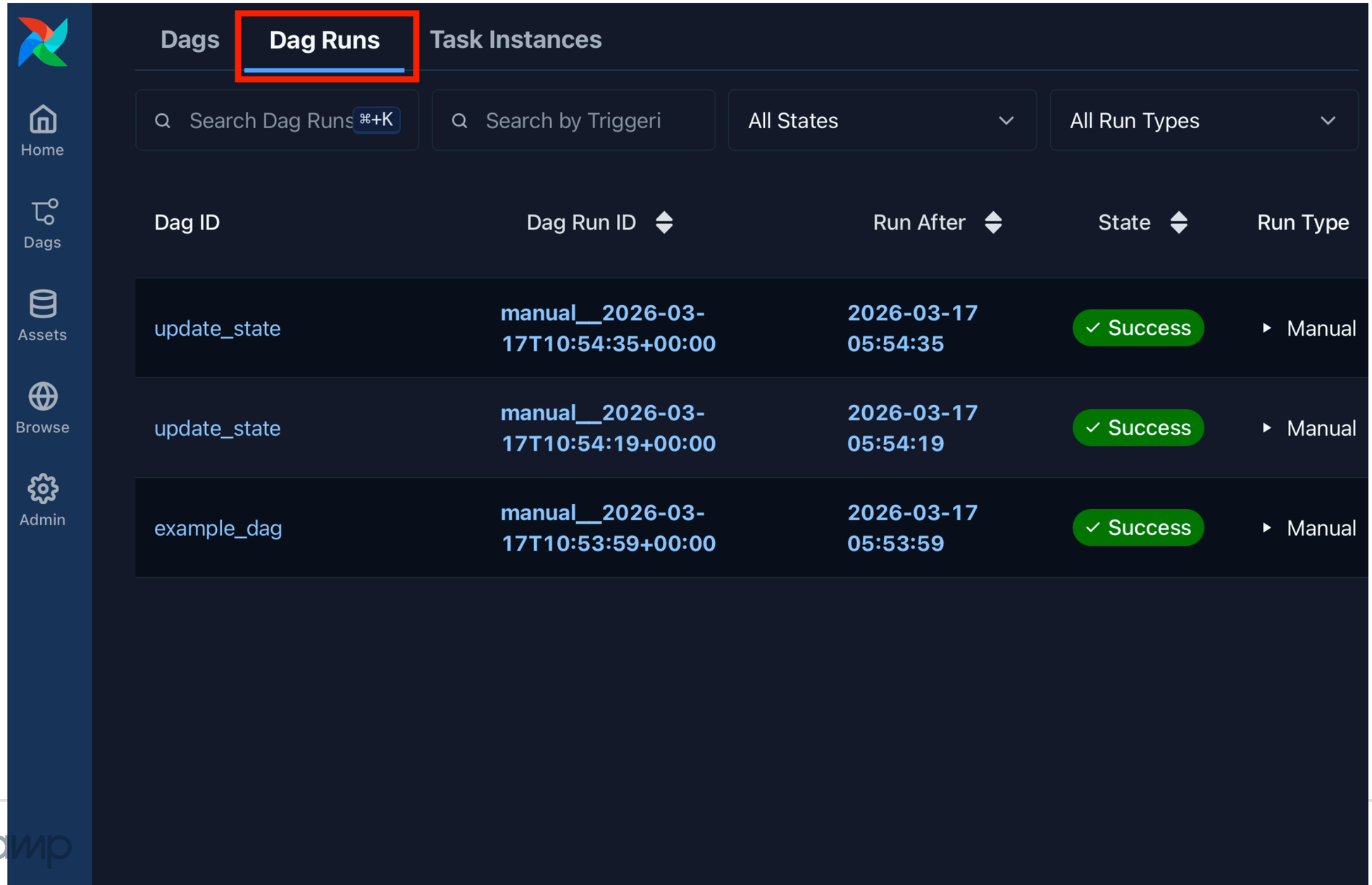
Dags view update_state (Graph)



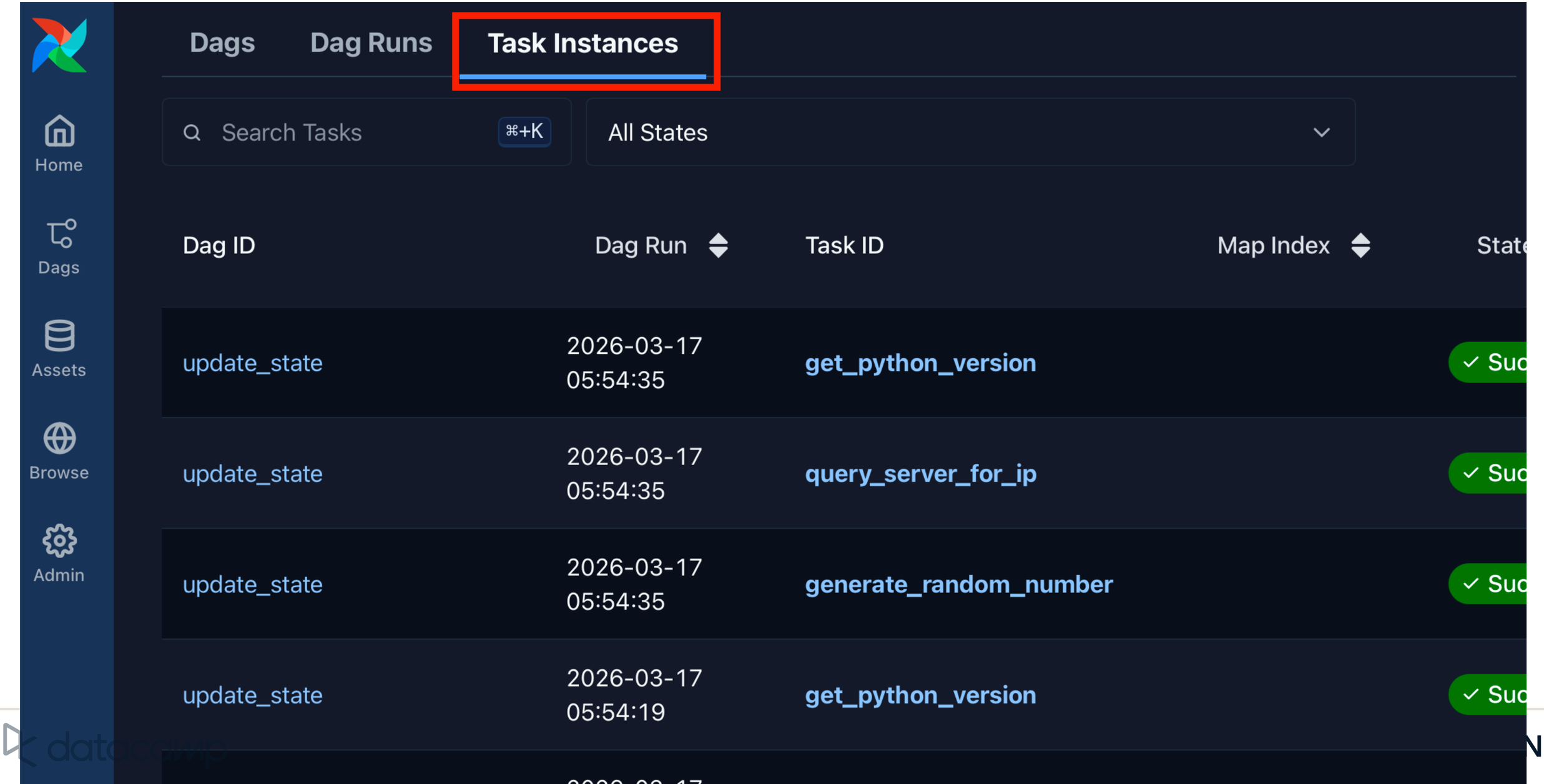
Dags view update_state (Code)



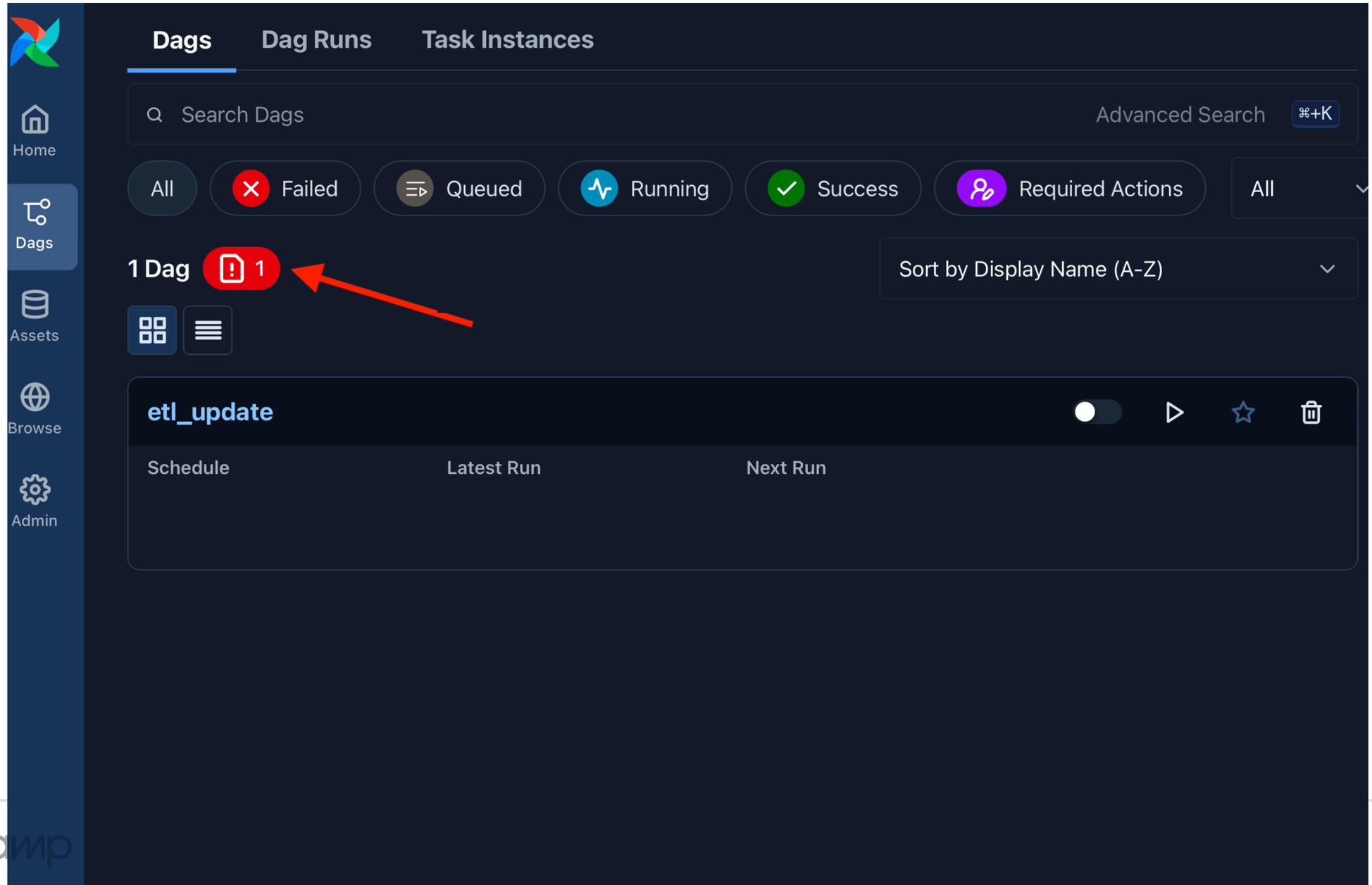
Dag Runs view



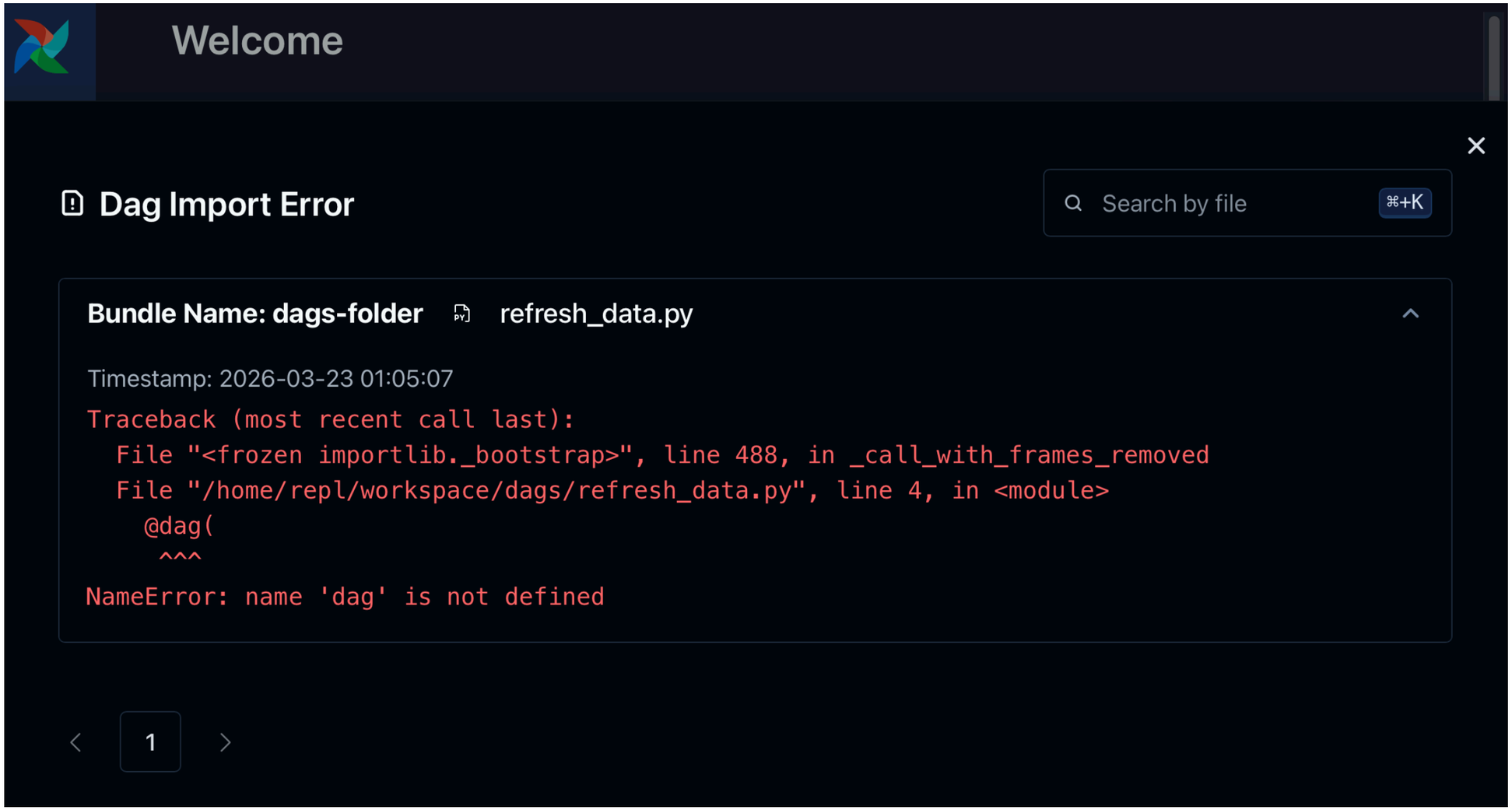
Task instances



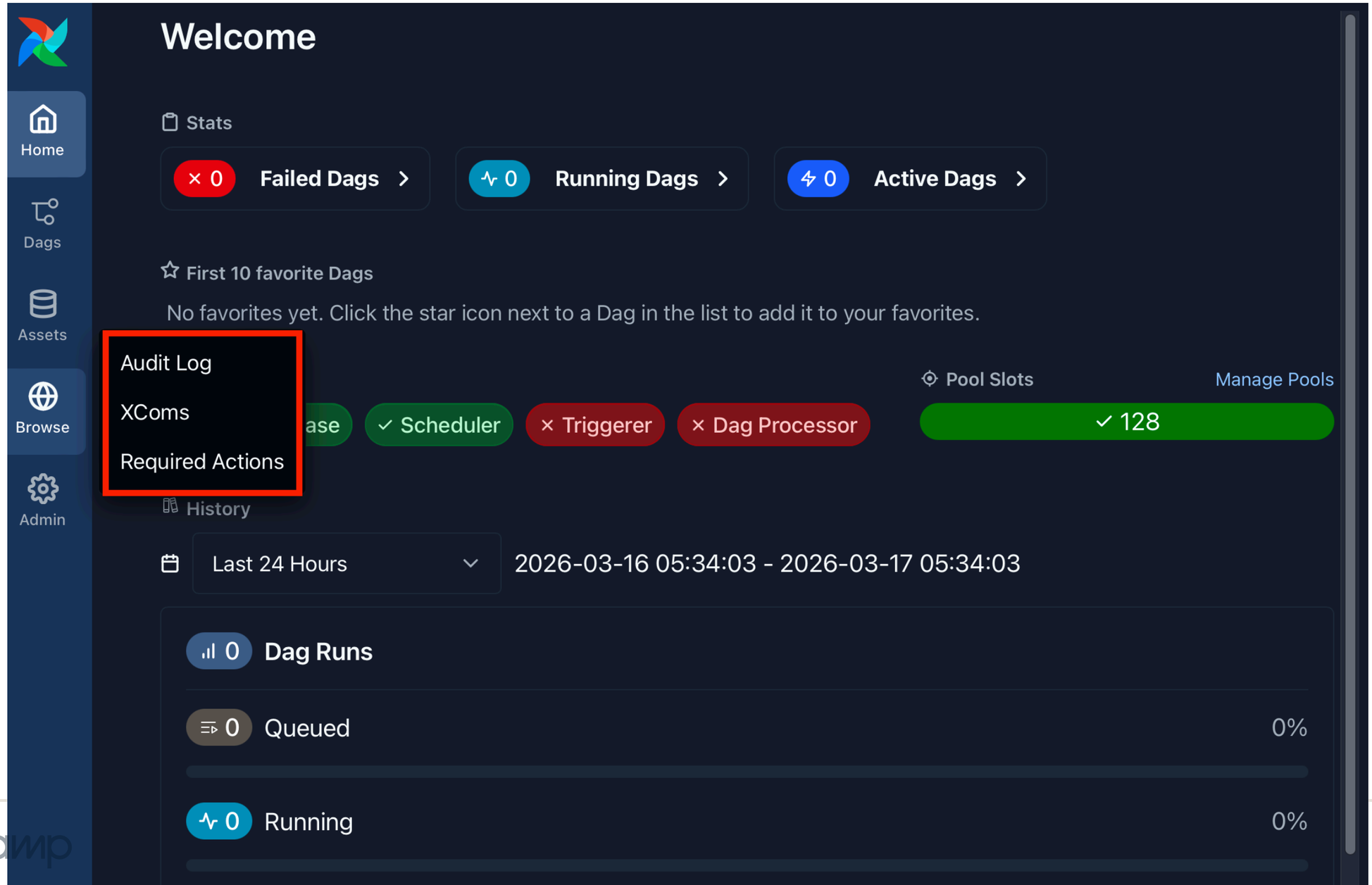
Dag Import Errors



Dag Import Errors Detail



Audit Log


Welcome
Home
Dags
Assets
Browse
Admin
Stats
0 Failed Dags
0 Running Dags
0 Active Dags
First 10 favorite Dags
No favorites yet. Click the star icon next to a Dag in the list to add it to your favorites.
Audit Log
XComs
Required Actions
Scheduler
Triggerer
Dag Processor
Pool Slots
Manage Pools
128
History
Last 24 Hours
2026-03-16 05:34:03 - 2026-03-17 05:34:03
0 Dag Runs
0 Queued
0%
0 Running
0%

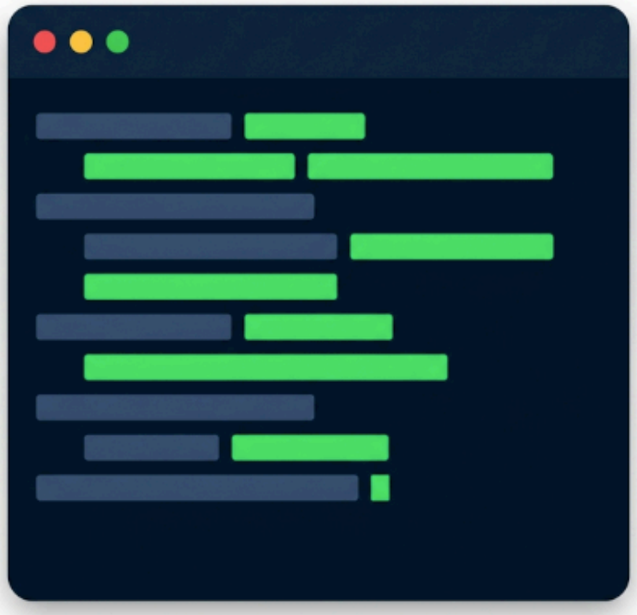
Audit Log


Audit Log
Filter
Home
Dags
Assets
Browse
Admin
When
Event
User
Extra
Dag ID
Run ID
Tas
2026-03-17 05:28:49
cli_dag_reserialize
repl
{...}
2 items
2026-03-17 05:28:47
cli_dag_reserialize
repl
{...}
2 items
2026-03-17 05:28:41
cli_scheduler
repl
{...}
2 items
2026-03-17 05:28:41
cli_api_server
repl
{...}
2 items

Web UI vs command line

- Equally powerful depending on needs
- Both communicate with the Airflow API-Server
- Web UI is often easier
- Command line tool may be easier to access depending on settings





Let's practice!

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